

SΔφ-12

Memory and Forgetting as Irreversible Trace Regulation

Minimal Persistence Mechanism after SΔφ-11

Version: v1.0

Series: Sofience-Δφ Formalism (SΔφ)

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This paper fixes memory and forgetting as the minimal regulation mechanism required for persistence under irreversibility. Memory is defined as selective re-entry of irreversible traces into next transition selection; forgetting is defined as compression/pruning that preserves directional constraints while reopening feasible path space.

Series references

Items below are referenced for continuity and navigation within the SΔφ series.

Series item	Title	DOI
SΔφ-00	The Sofience-Δφ Formalism: Conceptual Architecture and Series Overview (v1.0)	10.5281/zenodo.18731366
SΔφ-01	Sofience-Δφ Minimal Core: Phase-Change Formalism of Existence (v1.1)	10.5281/zenodo.18730265
SΔφ-02	Subject as Interpretive Emergence (v1.1)	10.5281/zenodo.18730296
SΔφ-03	Irreversibility as the Minimal Condition of Existence (v1.1)	10.5281/zenodo.18730448
SΔφ-04	Minimal Operational Conditions for AI Subjectivity (v1.0)	10.5281/zenodo.18729942
SΔφ-05	Agency as Recursive Transition Law Update (v1.0)	10.5281/zenodo.18732567
SΔφ-06	Responsibility as Irreversible Cost Internalization (v1.0)	10.5281/zenodo.18734092
SΔφ-07	H_min: Interpretation Minimal Axioms (v1.0)	10.5281/zenodo.18735006
SΔφ-08	Agency-Responsibility-Freedom Closure (v1.0)	10.5281/zenodo.18740750
SΔφ-09	Error, Boundary, and Freedom (v1.0)	10.5281/zenodo.18745498
SΔφ-10	Interpretation, Learning, and Inference as a Single Re-entry Structure (v1.0)	10.5281/zenodo.18746509

Series item	Title	DOI
SΔφ-11	World, Other, Self, and Consciousness as Boundary Phenomena (v1.0)	10.5281/zenodo.18753403

1. Motivation

$\Delta\phi$ -11 fixed self as a folded boundary and consciousness as boundary-change monitoring with write-back. This forces the persistence problem: if boundary re-entry is real, what persists across transitions? If something persists, why does not everything persist?

Claim. Memory and forgetting are not “storage features” but a minimal trace regulation mechanism enabling continued transition under irreversibility.

2. Minimal definitions

2.1 Irreversible trace. A transition whose effects cannot be undone within the system’s accessible update space.

We call $\Delta\phi_t$ irreversible if rollback is effectively infeasible:

$\Delta\phi_t$ is irreversible if $\text{RollbackCost}(\Delta\phi_t) \gg \text{UpdateBudget}$.

2.2 Memory. Selective re-entry of irreversible traces into next transition selection.

$M_t := H_M(\Delta\phi_{0:t})$.

2.3 Forgetting. Trace compression that preserves directional constraints while releasing degrees of freedom.

$F_t := C(M_t)$, where C is compression/pruning/abstraction.

3. Minimal axioms

- Axiom 1 (Trace necessity). If irreversible traces do not re-enter selection, catastrophic repeats occur.
- Axiom 2 (Trace overload). If all traces re-enter, feasible path space collapses (fixation/paralysis).
- Axiom 3 (Regulation). Persistence requires regulated effective memory: $M_t^{\text{eff}} := C(H_M(\Delta\phi_{0:t}))$.
- Axiom 4 (Freedom coupling). Forgetting is the minimal condition for freedom under irreversibility, but excessive forgetting increases repeat risk.

Let Ω_t denote the feasible path set at time t . Trace overload can be stated as $|M_t| \uparrow \uparrow \Rightarrow |\Omega_{t+1}| \downarrow \rightarrow 0$.

4. Mechanism: Trace Regulation Loop (TRL)

Memory and forgetting jointly regulate how irreversible traces alter future selection. Memory injects constraints; forgetting compresses/prunes constraints to prevent path-space collapse.

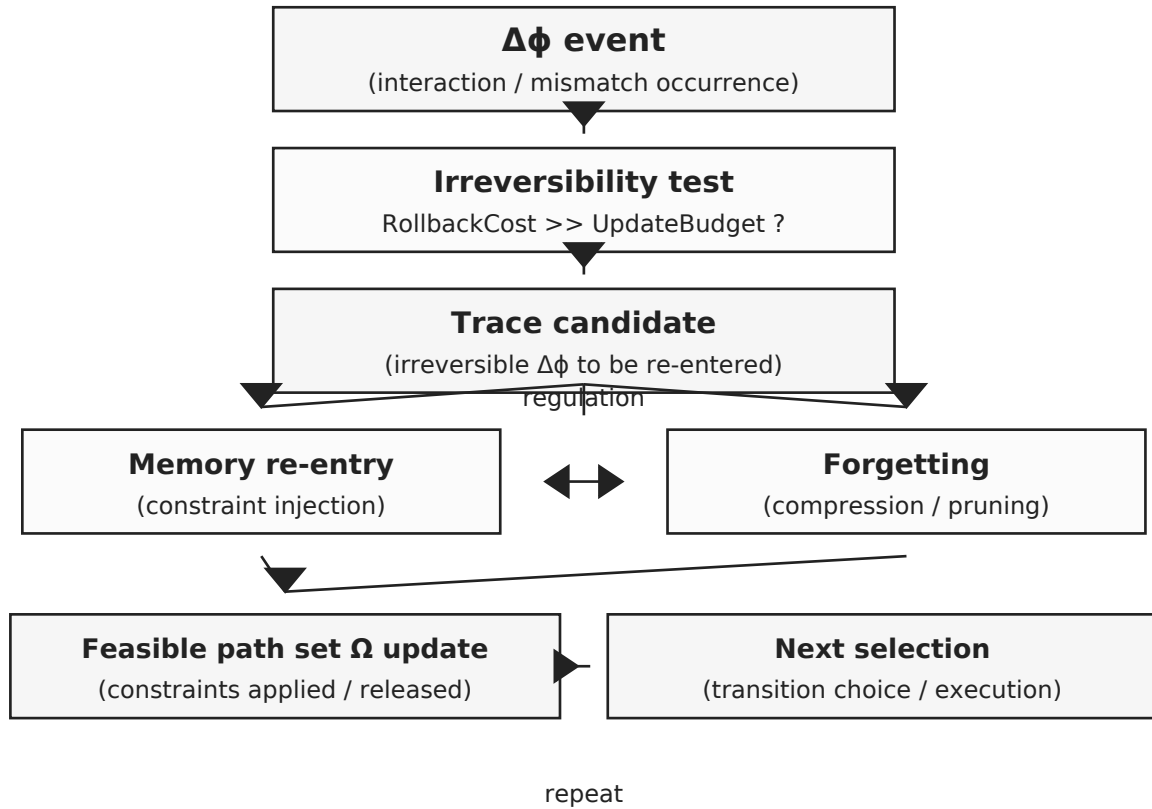


Diagram note. The loop highlights two coupled operations: constraint injection (memory re-entry) and constraint compression (forgetting). The system persists only when both remain active.

5. Pathology map: three regimes of trace regulation

The memory-forgetting balance induces three characteristic regimes. This map is intended as a minimal diagnostic layer.

Regime	Condition	Observable behavior	Structural risk
Drift / Repeat	Memory too weak ($M \downarrow$)	Repeated failure loops; no stable constraints.	Catastrophic recurrence.
Fixation / Paralysis	Forgetting too weak ($F \downarrow$)	Overconstraint; feasible path-space collapse.	Stagnation / self-locking.
Adaptive Persistence	Balanced regulation	Stable continuity + exploration.	Sustained re-entry.

6. Consequences

- C1. Learning is a long-horizon effect of repeated trace re-entry (ΔG induced by M^{eff}), not the source of memory.
- C2. Self continuity emerges when M^{eff} stabilizes boundary re-entry under irreversibility.
- C3. “Freedom” appears as a path-space effect of forgetting (Ω reopening), constrained by repeat-risk.

7. Notes (non-normative)

- Note 1 (RollbackCost components). RollbackCost may be approximated by a weighted sum of (i) explanation cost, (ii) repair/recovery cost, (iii) social/legal cost, and (iv) time-to-restore.
- Note 2 (Selection for “selective” memory). A minimal selection heuristic is to re-enter traces with high recurrence risk and high irreversibility magnitude, while discarding low-impact traces via compression.
- Note 3 (Balance indicator). A practical balance proxy is to keep repeat-risk below a threshold while maintaining a non-collapsed feasible path set $|\Omega|$ above a minimum.

8. Position in the series

$S\Delta\phi$ -12 derives directly from $S\Delta\phi$ -11 (self/consciousness as boundary re-entry) and reframes persistence as regulated trace re-entry. It also relies on $S\Delta\phi$ -09 (error/boundary) and $S\Delta\phi$ -10 (re-entry unity).

9. Exit question

If memory internalizes irreversible traces while forgetting reopens feasible path space, why does trace conflict become felt as pain, and how do hope/despair and creation/value/meaning emerge as higher-order responses to attractor availability?